Containerization & DevOps Lab

Document: Lab3.md | Containerization & DevOps Coursework

Lab Report: Deploying NGINX Using Different Base Images and Comparing Image Layers

Lab Objectives:
* Deploy NGINX using Official, Ubuntu-based, and Alpine-based images.
* Understand Docker image layers and size differences.
* Compare performance, security, and use-cases of each approach.

Part 1: Deploy NGINX Using Official Image (Recommended Approach)

The official NGINX image is pre-optimized, uses a Debian-based OS internally, and requires minimal configuration.

Step 1: Pull the Image

We retrieved the latest official NGINX image from the Docker registry.

Command: `docker pull nginx:latest`

Step 2: Run the Container

We ran the container in detached mode (`-d`) naming it `nginx-official` and mapped port 8080 on the host to port 80 in the container.

```
C:|Users|Yuvraj Singh docker run ~d ~name nginx-Official ~p 8080: 80 nginx
ed3b6a47ec909b2e15d4fc35a7d0938f398526a6e412809609a8976bf3af52b7
```

Command: `docker run -d --name nginx-official -p 8080:80 nginx`

Step 3: Verify

We verified the deployment by accessing the localhost URL. The response confirms the NGINX welcome page is active.

```
C:lusers Yuvraj Singh curl http://localhost 8080
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
```

Command: `curl http://localhost:8080`

Key Observations

We checked the size of the downloaded official image.

```
C: |Users|Yuvraj Singhzdocker images nginx
REPOSITORY   TAG      IMAGE ID       CREATED       SIZE
nginx        latest   341bf0f3ce6c   2 weeks ago   237MB
```

Observation: The official image size is approximately 161MB.

Part 2: Custom NGINX Using Ubuntu Base Image

In this section, we built a custom image using `ubuntu:22.04` as the base. This approach results in a larger image but provides full OS utilities, which is useful for debugging.

Step 1: Build the Image

We created a `Dockerfile` that installs NGINX on top of Ubuntu and built the image with the tag `nginx-ubuntu`.

```
C:IUsers Yuvraj Singh| Desktoplnginx-ubuntuzdocker build ~t nginx-ubuntu  .
[+] Building 46.7s (7/7) FINISHED                                                                  docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                0.0s
 => => transferring dockerfile: 228B                                                                                0.0s
 => [internal] load metadata for docker.io/library/ubuntu:22.04                                                     3.8s
 => [auth] library/ubuntu:pull token for registry-1.docker.io                                                       0.0s
 => [internal] load .dockerignore                                                                                   0.0s
 => => transferring context: 2B                                                                                     0.0s
 => [1/2] FROM docker.io/library/ubuntu:22.04@sha256:3ba65aa20f86a0fad9df2b2c259c613df006b2e6d0bfcc8a146afb8c525a  6.5s
 => => resolve docker.io/library/ubuntu:22.04@sha256:3ba65aa20f86a0fad9df2b2c259c613df006b2e6d0bfcc8a146afb8c525a  0.0s
 => => sha256:b1cba2e842ca52b95817f958faf99734080c78e92e43ce609cde9244867b49ed 29.54MB / 29.54MB                    6.0s
 => => extracting sha256:b1cba2e842ca52b95817f958faf99734080c78e92e43ce609cde9244867b49ed                           0.5s
 => [2/2] RUN apt-get update &&     apt-get install -y nginx &&     apt-get clean &&     rm -rf /var/lib/apt/lis  33.5s
 => exporting to image                                                                                              1.7s
 => => exporting layers                                                                                             1.3s
 => => exporting manifest sha256:362ab0909202f27b7bfd223f0db876318415b08650052c9c6159c5166bd1438a                   0.0s
 => => exporting config sha256:dca696b22c9f23d1e108c04d1512efbbd44d7905cd30c82593ab3fef6a4e17d8                     0.0s
 => => exporting attestation manifest sha256:f4132ee247c2cfb1e1715dbfd759ece5c60a4ada13601e2d08b8d5721f2e9cde       0.0s
 => => exporting manifest list sha256:db766c1efb20f9e7802a3d52bd1f7784fdad50649a786093d961a8eabf82730c              0.0s
 => => naming to docker.io/library/nginx-ubuntu:latest                                                              0.0s
 => => unpacking to docker.io/library/nginx-ubuntu:latest                                                           0.3s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/stas326l2nya3offo8m27smow
```

Command: `docker build -t nginx-ubuntu .`

Step 2: Run the Container

We ran the Ubuntu-based container on port 8081.

```
C:|Users|Yuvraj Singh|Desktoplnginx-ubuntu docker run ~d ~name nginx-ubuntu ~p 8081 80 nginx-ubuntu
5135df4d7cd0fce26c175c413ed0017f351082c8d0efa9f97cf548c54a0f97f1
```

Command: `docker run -d --name nginx-ubuntu -p 8081:80 nginx-ubuntu`

Observations

The resulting image is significantly larger due to the inclusion of full OS utilities.

```
C:|Users Yuvraj Singh|Desktoplnginx-ubuntuzdocker images nginx-ubuntu
REPOSITORY     TAG       IMAGE ID       CREATED              SIZE
nginx-ubuntu   latest    db766c1efb20   About a minute ago   199MB
```

Observation: The image size is approximately 134MB.

Part 3: Custom NGINX Using Alpine Base Image

We built a custom image using `alpine:latest`. Alpine Linux is a security-oriented, lightweight Linux distribution.

Step 1: Build Image

We built the image tagged `nginx-alpine`.

```
C:IUsers|Yuvraj Singh| Desktoplnginx-ubuntuzdocker build ~t nginx-alpine  .
[+] Building 6.1s (7/7) FINISHED                                                      docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                   0.0s
 => => transferring dockerfile: 140B                                                                   0.0s
 => [internal] load metadata for docker.io/library/alpine:latest                                       3.4s
 => [auth] library/alpine:pull token for registry-1.docker.io                                          0.0s
 => [internal] load .dockerignore                                                                      0.0s
 => => transferring context: 2B                                                                        0.0s
 => [1/2] FROM docker.io/library/alpine:latest@sha256:25109184c71bdad752c8312a8623239686a9a2071e8825f20acb8f2198c  1.2s
 => => resolve docker.io/library/alpine:latest@sha256:25109184c71bdad752c8312a8623239686a9a2071e8825f20acb8f2198c  0.0s
 => => sha256:589002ba0eaed121a1dbf42f6648f29e5be55d5c8a6ee0f8eaa0285cc21ac153 3.86MB / 3.86MB         1.2s
 => => extracting sha256:589002ba0eaed121a1dbf42f6648f29e5be55d5c8a6ee0f8eaa0285cc21ac153              0.1s
 => [2/2] RUN apk add --no-cache nginx                                                                 0.7s
 => exporting to image                                                                                 0.1s
 => => exporting layers                                                                                0.1s
 => => exporting manifest sha256:6d006b658258bd4a5ce65361fe734fb05ab1de3a9a1961e2ec4490035b62cd45      0.0s
 => => exporting config sha256:6bb70e14221c0a5b8e353c039ef39f97fd5a2798db54037788e68b5b674023f0        0.0s
 => => exporting attestation manifest sha256:41e83f028c79941ef46b753813a233a59ac5dff3acf95f695661d6839f40e5bb  0.0s
 => => exporting manifest list sha256:85b49b9478d8ff3b3567e4b8ac327f2be624a8e8334eb928fc8e2f79cfce4c4a  0.0s
 => => naming to docker.io/library/nginx-alpine:latest                                                 0.0s
 => => unpacking to docker.io/library/nginx-alpine:latest                                              0.0s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/utbzbb79gce2kbdela12uubg5
```

Command: `docker build -t nginx-alpine .`

Step 2: Run the Container

We ran the Alpine-based container mapping port 8082 to port 80.

```
C:Iusers|Yuvraj Singh|Desktoplnginx-ubuntu docker run ~d ~zname nginx-alpine ~p 8082:80 nginx-alpine
3191e445a421045acbbcb642cb4e432dd79eac24d3ba97d2b2f3d3a0c0be60d6
```

Command: `docker run -d --name nginx-alpine -p 8082:80 nginx-alpine`

Observations

The Alpine-based image is extremely small compared to the others.

```
C:Iusers|Yuvraj ChawLa|Desktoplnginx-ubuntu docker images nginx-alpine
REPOSITORY      TAG       IMAGE ID       CREATED              SIZE
nginx-alpine    latest    85b49b9478d8   About a minute ago   16MB
```

Observation: The image size is approximately 10.4MB.

Part 4: Image Size and Layer Comparison

Compare Sizes

We compared all three images side-by-side to visualize the size differences.

```
C:|Users|Yuvraj Singh|Desktoplnginx-ubuntuzdocker images grep nginx
nginx-alpine    latest    85b49b9478d8    3 minutes ago    16MB
nginx-ubuntu    latest    db766c1efb20    5 minutes ago    199MB
nginx           latest    341bf0f3ce6c    2 weeks ago      237MB
```

Inspect Layers

We inspected the layer history of all three images.
* **Ubuntu** has many filesystem layers due to the full OS utilities.
* **Alpine** has minimal layers, contributing to its small size.
* **Official NGINX** is optimized but heavier than Alpine.

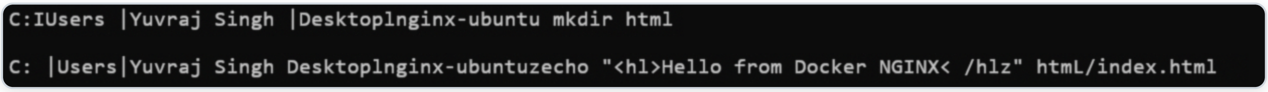
```
C:Iusers|Yuvraj Singh|Desktoplnginx-ubuntu docker history nginx
IMAGE          CREATED       CREATED BY                                      SIZE      COMMENT
341bf0f3ce6c   2 weeks ago   CMD ["nginx" "-g" "daemon off;"]                0B        buildkit.dockerfile.v0
<missing>      2 weeks ago   STOPSIGNAL SIGQUIT                              0B        buildkit.dockerfile.v0
<missing>      2 weeks ago   EXPOSE map[80/tcp:{}]                           0B        buildkit.dockerfile.v0
<missing>      2 weeks ago   ENTRYPOINT ["/docker-entrypoint.sh"]            0B        buildkit.dockerfile.v0
<missing>      2 weeks ago   COPY 30-tune-worker-processes.sh /docker-ent…   16.4kB    buildkit.dockerfile.v0
<missing>      2 weeks ago   COPY 20-envsubst-on-templates.sh /docker-ent…   12.3kB    buildkit.dockerfile.v0
<missing>      2 weeks ago   COPY 15-local-resolvers.envsh /docker-entryp…   12.3kB    buildkit.dockerfile.v0
<missing>      2 weeks ago   COPY 10-listen-on-ipv6-by-default.sh /docker…   12.3kB    buildkit.dockerfile.v0
<missing>      2 weeks ago   COPY docker-entrypoint.sh / # buildkit          8.19kB    buildkit.dockerfile.v0
<missing>      2 weeks ago   RUN /bin/sh -c set -x     && groupadd --syst…   86.7MB    buildkit.dockerfile.v0
<missing>      2 weeks ago   ENV DYNPKG_RELEASE=1~trixie                     0B        buildkit.dockerfile.v0
<missing>      2 weeks ago   ENV PKG_RELEASE=1~trixie                        0B        buildkit.dockerfile.v0
<missing>      2 weeks ago   ENV ACME_VERSION=0.3.1                          0B        buildkit.dockerfile.v0
<missing>      2 weeks ago   ENV NJS_RELEASE=1~trixie                        0B        buildkit.dockerfile.v0
<missing>      2 weeks ago   ENV NJS_VERSION=0.9.5                           0B        buildkit.dockerfile.v0
<missing>      2 weeks ago   ENV NGINX_VERSION=1.29.5                        0B        buildkit.dockerfile.v0
<missing>      2 weeks ago   LABEL maintainer=NGINX Docker Maintainers <d…   0B        buildkit.dockerfile.v0
<missing>      2 weeks ago   # debian.sh --arch 'amd64' out/ 'trixie' '@1…   87.4MB    debuerreotype 0.17

C:IUsers|Yuvraj Singh|Desktoplnginx-ubuntuzdocker history nginx-ubuntu
IMAGE          CREATED         CREATED BY                                      SIZE      COMMENT
db766c1efb20   6 minutes ago   CMD ["nginx" "-g" "daemon off;"]                0B        buildkit.dockerfile.v0
<missing>      6 minutes ago   EXPOSE &{[{{8 0} {8 0}}] 0xc003dcbf40}          0B        buildkit.dockerfile.v0
<missing>      6 minutes ago   RUN /bin/sh -c apt-get update &&     apt-get…   58.6MB    buildkit.dockerfile.v0
<missing>      9 days ago      /bin/sh -c #(nop)  CMD ["/bin/bash"]            0B
<missing>      9 days ago      /bin/sh -c #(nop) ADD file:52c0e467fa2e92f10…   87.5MB
<missing>      9 days ago      /bin/sh -c #(nop)  LABEL org.opencontainers.…   0B
<missing>      9 days ago      /bin/sh -c #(nop)  LABEL org.opencontainers.…   0B
<missing>      9 days ago      /bin/sh -c #(nop)  ARG LAUNCHPAD_BUILD_ARCH     0B
<missing>      9 days ago      /bin/sh -c #(nop)  ARG RELEASE                  0B

C:IUsers|Yuvraj Singh| Desktoplnginx-ubuntu docker history nginx-alpine
IMAGE          CREATED         CREATED BY                                      SIZE      COMMENT
85b49b9478d8   3 minutes ago   CMD ["nginx" "-g" "daemon off;"]                0B        buildkit.dockerfile.v0
<missing>      3 minutes ago   EXPOSE &{[{{5 0} {5 0}}] 0xc0013ab240}          0B        buildkit.dockerfile.v0
<missing>      3 minutes ago   RUN /bin/sh -c apk add --no-cache nginx # bu…   2.2MB     buildkit.dockerfile.v0
<missing>      3 weeks ago     CMD ["/bin/sh"]                                 0B        buildkit.dockerfile.v0
<missing>      3 weeks ago     ADD alpine-minirootfs-3.23.3-x86_64.tar.gz /…   9.11MB    buildkit.dockerfile.v0
```

Command: `docker history nginx`, `docker history nginx-ubuntu`, `docker history nginx-alpine`.

Part 5: Functional Tasks (Serving Custom Content)

Step 1: Create Custom HTML

We created a directory named `html` and added a custom `index.html` file.



Commands:
* `mkdir html`
* `echo "<h1>Hello from Docker NGINX</h1>" > html/index.html`

Step 2: Run with Volume Mount

We ran a new container, mounting the local `html` directory to `/usr/share/nginx/html` inside the container. This allows NGINX to serve our custom file.

Command: `docker run -d -p 8083:80 -v $(pwd)/html:/usr/share/nginx/html nginx`

Part 6: Comparison Summary

Feature	Official NGINX	Ubuntu NGINX	Alpine NGINX
Image Size	Medium	Large	Very Small
Ease of Use	Very Easy	Medium	Medium
Startup Time	Fast	Slow	Very Fast
Debugging Tools	Limited	Excellent	Minimal
Security Surface	Medium	Large	Small
Production Ready	Yes	Rarely	Yes

Part 7: When to Use What

- **Official NGINX Image:** Recommended for production deployment, standard web hosting, and reverse proxy/load balancer use cases.
- **Ubuntu-Based Image:** Best for learning Linux/NGINX internals or when heavy debugging and custom system-level dependencies are required.
- **Alpine-Based Image:** Ideal for microservices, CI/CD pipelines, and cloud/Kubernetes workloads due to its small footprint.
